

Research Statement

Md Abu Sayed

Postdoctoral Researcher, Center for Digital Cardiovascular Innovations, University of Miami

1. Research Vision

In many safety-critical settings, a decision has to be made before the full picture is available. A naval operator has to judge whether an approaching vessel is hostile from a short segment of its track. An interventional cardiologist has to choose how to prepare a calcified lesion before knowing how the stent will expand. My research develops machine learning methods for these situations: models that infer the state and likely evolution of a system from partial, noisy, or weakly labeled observations, and that report their predictions in a form people can check and act on. The central question of my work is:

How can learning systems predict behavior and outcomes from incomplete observations, and deliver predictions that are accurate, calibrated, and interpretable enough to support safety-critical decisions?

I study this question with temporal and generative deep learning, simulation-based supervision, and evaluation under noise and uncertainty. I have developed and tested these methods in two domains: **multi-agent intent recognition** for autonomous systems, the focus of my Ph.D. research funded by the Office of Naval Research (ONR), and **cardiovascular image analysis and clinical decision support**, the focus of my postdoctoral work at the University of Miami Miller School of Medicine.

2. Joint Intent Recognition and Trajectory Prediction

Maritime surveillance requires two answers about every encounter: which behavior is taking place, and where each vessel will be in the next few seconds. I worked on this problem in an ONR program in collaboration with Huntington Ingalls Industries (HII), which will deploy the resulting models on water as part of the project. The data are two-vessel encounters simulated by HII across seven behaviors (non-hostile benign, cross, head-on, and overtake; hostile block, herd, and ram) at three levels of sensor noise, with about 8,000 scenarios and 880,000 training windows per noise level. I also led most of the development of NavySim, the program’s Unity multi-vessel simulator for scenario design, local testing, and visualization (IEEE CoG 2024; *IEEE Transactions on Games* 2026), and developed ThreatMap, a real-time risk map that combines sensor coverage, vulnerability, and closest-point-of-approach kinematics (HMS 2024; M.S. thesis).

Earlier stages. I first addressed scenario classification from long observation windows, reaching about 97% seven-class accuracy with feature-aware LSTM and Transformer models (CASE 2025; FMLDS 2025). I then compared six Kalman-filter variants for trajectory prediction. They forecast routine traffic well, but they have no notion of intent, and their errors grow sharply on hostile maneuvers.

MTITP. The final contribution of my dissertation is MTITP (Multi-Task Intent and Trajectory Prediction), a single model that answers both questions. A shared encoder processes three input streams (per-vessel kinematics, pairwise geometry, and symbolic COLREGs features) and feeds three heads: current intent, per-step future intent, and an intent-conditioned distribution over future trajectories. The model observes 20 s of an encounter and predicts 5 s ahead. Ablations show that the two tasks support each other. Removing the intent heads increases trajectory error by 1.25 m, the largest effect in the study, while removing the geometric and symbolic streams lowers future-intent mAP from 99.4% to 66.2% with no change in trajectory error. For adversarial training, I designed MarITGAN, a critic that evaluates generated paths against the predicted intent and the pairwise geometry of the two vessels. With the backbone, data, and generator unchanged, MarITGAN lowers displacement error by 35% compared with a generic WGAN-GP critic and preserves multi-modal sampling.

Table 1 summarizes the results. MTITP reaches 99.4% future-intent mAP, reduces displacement error by a factor of 2.3 relative to the best Kalman filter, and improves seven-class accuracy by ten points over the strongest published classifier on the same noisy data. Under heavy noise, intent accuracy drops only to 97.2% while trajectory error

Table 1: MTITP and baselines on HII-simulated encounters (7 behaviors; 20-s observation, 5-s prediction).

Model	Data	Intent accuracy (7-class)	ADE / FDE (m)
Best Kalman filter (EKF-CTRV)	Noiseless	n/a	5.06 / 10.52
MTITP, generic WGAN-GP critic	Noiseless	99.2%	3.42 / 5.97
MarITGAN (intent-aware critic)	Noiseless	99.2%	2.23 / 3.85
MTITP (L2 objective)	Noiseless	99.2%	2.17 / 3.69
Published Bi-LSTM classifier (CASE 2025)	Moderate noise	88.0%	n/a
MTITP (L2 objective)	Moderate noise	98.4%	7.38
MTITP (L2 objective)	Heavy noise	97.2%	30.20

risers with the noise level, which indicates that behavior recognition is much less sensitive to sensor noise than position forecasting. Kalman filters remain more accurate on routine non-hostile traffic, whereas MTITP performs best on the hostile herd and ram encounters, so a practical system would combine the two. Two first-author manuscripts have been submitted, one on the MTITP framework (*IEEE T-ITS*) and one on the MarITGAN critic (*Eng. Appl. of AI*).

Open problems. The HII data reproduce COLREGs geometry but not the artifacts of real AIS data, so validation on independently collected encounters is the next step. Other open problems are extension beyond two vessels with graph-based models, diffusion or flow-based generators for longer horizons, and online updating of intent as an encounter develops.

3. Cardiovascular Image Analysis and Clinical Decision Support

Intravascular imaging poses a similar problem in a clinical setting. An IVUS or OCT pullback images one artery with modality-specific noise, expert labels are costly and often disagree, and the outcome that matters most, how the lesion will respond to intervention, cannot be observed before the procedure. At the Center for Digital Cardiovascular Innovations, I work on three connected problems.

- **Segmentation under label noise.** Multi-class segmentation of lumen, vessel wall, and plaque composition (calcium, fibrous, fibrolipidic) across HD-IVUS and NIRS-IVUS, using more than 40,000 expert-labeled IVUS frames, with angiography, OCT, and OCT-NIRS as upcoming extensions. Because experts disagree on composition labels, I use inter-observer variability as the reference level for validating these classes. A manuscript on multi-class high-definition IVUS segmentation is under review at *Scientific Reports*, and a head-to-head comparison of intracoronary imaging modalities is under review at *JACC: Cardiovascular Interventions*.
- **From imaging to outcome.** Automated IVUS-based calcium scoring (OCT-based scoring upcoming) to predict stent under-expansion before intervention, so that lesion preparation can be planned in advance.
- **Decision support from simulation.** Procedural records contain only the outcome of the strategy that was used. To recommend a lesion-preparation strategy (non-compliant balloon, intravascular lithotripsy, or atherectomy) and a bifurcation stenting technique, we learn from a simulation-derived corpus of 1,200 lesion phenotypes and coronary anatomies, which provides outcomes for every strategy.

4. Common Methodology

The two lines of work share the same methodological problems (Table 2). In both, models predict from partial observations, must handle noisy inputs and inconsistent labels, and need outputs that a domain expert can inspect. Both also depend on simulation for training signals that real data cannot supply: adversarial encounters that cannot be staged at sea, and treatment strategies that were never performed on a given patient. How well models trained on simulation transfer to real systems is therefore a common open question and a central topic of my future research.

Table 2: Shared problems across the two research domains.

Problem	Autonomous systems	Cardiovascular imaging
Partial observation	Intent and future path from a 20-s window	Outcome from pre-procedure imaging
Noisy inputs	Sensor noise at three levels	Imaging artifacts; modality differences
Unreliable labels	Ambiguous behavior boundaries	Inter-observer variability in composition
Interpretability	ThreatMap risk maps; explicit intent outputs	Interpretable decision support for clinicians
Simulated supervision	HII-simulated encounters; NavySim testing	Simulated outcomes for each strategy

5. Research Program

My group will pursue three research thrusts. Each is designed to produce a student-led first paper within two years and to be fundable on its own.

Thrust 1: Joint intent and trajectory prediction for multi-agent systems. I will extend MTITP from two-vessel encounters to scenes with many agents, and from maritime traffic to pedestrians, road users, and people working alongside robots, where the same joint problem affects the safety of autonomous vehicles and collaborative robots. Graph-Transformer encoders will model interactions, and conditional diffusion or CVAE decoders will produce multiple calibrated futures. A central question is how early intent can be inferred at a specified level of risk, which I will address with conformal and selective prediction. This work has already started: I initiated and supervise an M.S. thesis on pedestrian intent recognition for autonomous vehicles at Khulna University.

Thrust 2: Reliable intravascular image analysis for outcome prediction. I will develop segmentation and outcome models that are robust to label noise and calibrated against inter-observer variability, move from frame-level segmentation to pullback-level temporal models that predict stent expansion and procedural risk, and study transfer across IVUS, OCT, and hybrid modalities. I initiated and supervise a B.S. thesis on coronary OCT segmentation that starts this thrust outside my current lab.

Thrust 3: Simulation-based decision support. I will study when a model trained on simulated outcomes can be trusted in the real system, and how the gap between simulation and reality should be measured and reported. The test beds are NavySim for autonomy and biomechanical simulation corpora for intervention planning. User studies will measure whether interpretable model outputs change the decisions of operators and clinicians.

Independence. The MTITP and MarITGAN approaches in my two submitted manuscripts are my own and were developed independently within my dissertation topic. I also initiated and supervise two thesis projects outside my current lab. Thrusts 1 and 3 do not depend on data from my current lab, and Thrust 2 builds on clinical collaborations that I will maintain and establish with local cardiology partners.

6. Funding Strategy and Fit

- **Years 1–2:** NSF CISE CRII (Thrust 1); NIH R21 through NHLBI or NIBIB with a clinical co-investigator (Thrust 2); ONR Young Investigator Program in maritime autonomy (Thrusts 1 and 3); NSF REU supplements for undergraduate researchers.
- **Years 3–5:** NSF CAREER (Thrusts 1 and 3); NSF–NIH Smart Health and Biomedical Research in the Era of AI (SCH) and NIH R01 with clinical partners (Thrust 2); NSF IIS Robust Intelligence and Cyber-Physical Systems.
- **Ongoing:** partnerships with intravascular imaging manufacturers, and international collaboration through my thesis supervision at Khulna University.

7. Long-Term Vision

My long-term goal is to build a research group that develops reliable predictive models for safety-critical decisions in autonomy and medicine. The group will release its simulators, benchmarks, and code, and train students to work across machine learning, simulation, and an application domain. I will judge the work by its accuracy and by whether it helps operators and clinicians decide better.